225-226页2,5; 250页3,6,8,10,11,12.
269页1,2,3.

225-2

Let $T \in \mathcal{L}(\mathbb{R}^3)$ be

$$\begin{pmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{pmatrix}$$

Show that there is a diagonalizable $D \in \mathcal{L}(\mathbb{R}^3)$ and a nilpotent $N \in \mathcal{L}(\mathbb{R}^3)$ such that $T = D + N$ and $DN = ND$.
Solution:

$$D = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -2 & 2 & 2 \end{pmatrix}, N = \begin{pmatrix} 2 & 0 & -1 \\ 2 & 0 & -1 \\ 4 & 0 & -2 \end{pmatrix}.$$

225-5

Let V be finite dim over $\mathbb{C}$, $T \in \mathcal{L}(V)$ and D be the diagonalizable part of T. Prove that if $g \in \mathbb{C}[x]$, then the diagonalizable part of $g(T)$ is $g(D)$.
Proof: Suppose $T = D + N$, then $g(T) - g(D) = N \cdot A$ where A is a polynomial in D, N. Since $DN = ND$, A commutes with N, hence $g(T) - g(D)$ is nilpotent, and $g(T) - g(D)$ commutes with $g(D)$ since they are both polynomials in D, N.

250-3

If $A \in \mathbb{C}^{5\times 5}$ with characteristic polynomial $f = (x-2)^3(x+7)^2$, and minimal polynomial $p = (x-2)^2(x+7)$, what is the Jordan form for A?
Solution: The elementary divisors are $p_1 = (x-2)^2(x+7)$, $p_2 = (x-2)(x+7)$. Hence the Jordan form is

$$A = \begin{pmatrix} 2 & & & & \\ 1 & 2 & & & \\ & & 2 & & \\ & & & 7 & \\ & & & & 7 \end{pmatrix}$$

250-8

Classify up to similarity all $A \in \mathbb{C}^{3\times 3}$ such that $A^3 = I$.
Proof: 10 diagonal matrices: $I, \omega I, \omega^2 I$,
$\mathrm{diag}(1,1,\omega), \mathrm{diag}(1,1,\omega^2), \mathrm{diag}(\omega,\omega,1), \mathrm{diag}(\omega,\omega,\omega^2), \mathrm{diag}(\omega^2,\omega^2,1), \mathrm{diag}(\omega^2,\omega^2,1)$, and $\mathrm{diag}(1,\omega,\omega^2)$.

250-10

Let n be a positive integer, $n \geqslant 2$, and let $N \in F^{n\times n}$ such that $N^n = 0$ but $N^{n-1} \neq 0$. Prove that N has no square root.
Proof: If $A^2 = N$, then $A^{2n} = 0$ implies $p_A \mid x^n$. Suppose $p_A = x^k$, then $A^k = 0$ implies $N^{\lfloor k/2 \rfloor} = 0$ a contradiction with $N^{n-1} \neq 0$.

250-11

Let $N_1, N_2 \in F^{6\times 6}$ be nilpotent. Suppose N_1, N_2 have the same minimal polynomial and the same nullity, prove that $N_1 \sim N_2$. Show that this is not true for $F^{7\times 7}$.
Proof: The nullity gives the number of Jordan blocks, and minimal polynomial gives the largest Jordan block.
$7 = 2+2+3 = 1+3+3$ have two partitions with same minimal polynomial and nullity.

269-1

If $N \in \mathcal{L}(V)$ is nilpotent, show that for any $f \in F[x]$ the semi-simple part of $f(N)$ is $c \cdot I$.
Proof: Suppose $f(x) = c + xg(x)$, then clearly $f(N) = c \cdot I + Ng(N)$ where $Ng(N)$ is nilpotent and commutes with $c \cdot I$.

269-2

Let $F \subset \mathbb{C}$, V finite dim over F, and $T \in \mathcal{L}(V)$ semi-simple. If $f \in F[x]$, prove that $f(T)$ is semi-simple.
Proof: T semisimple iff T diagonalizable over $\mathbb{C}$. If $P^{-1}TP$ is diagonal in $\mathbb{C}$, then $P^{-1}f(T)P$ is also diagonal, hence $f(T)$ is semi-simple.

269-3

Let $T \in \mathcal{L}(V)$, V finite dim over $F \subset \mathbb{C}$. Prove that T is semi-simple iff for any polynomial $f \in F[x]$ such that $f(T)$ is nilpotent, $f(T) = 0$.
Proof: If T is semi-simple then $f(T)$ is semi-simple so if $f(T)$ is nilpotent then $f(T) = 0$.
If T is not semi-simple, then its minimal polynomial p_A in $\mathbb{C}$ is not square-free. Suppose $p^2 \mid p_A$, let $f = p_A/p$, then $f(T)$ is nilpotent but $f(T) \neq 0$, a contradiction.